Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $\vec{a}$ and $\vec{b}$ are two vectors, such that $\vec{a} \cdot \vec{b} < 0$ and $|\vec{a} \cdot \vec{b}| = |\vec{a} \times \vec{b}|$, then the angle between vectors $\vec{a}$ and $\vec{b}$ is

(A) π

(B) $7\pi/4$

(C) $\pi/4$

(D) $3\pi/4$

Q.2

If $\vec{a}, \vec{b}$ and $\vec{c}$ are three mutually perpendicular vectors, then the vector which is equally inclined to these vectors

(A) $\vec{a} + \vec{b} + \vec{c}$

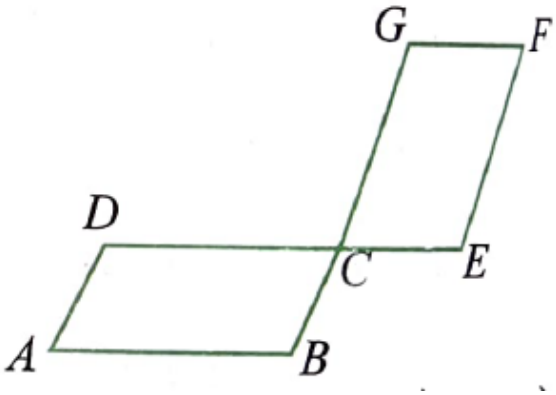
(B) $\frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|} + \frac{\vec{c}}{|\vec{c}|}$

(C) $\frac{\vec{a}}{|\vec{a}|^2} + \frac{\vec{b}}{|\vec{b}|^2} + \frac{\vec{c}}{|\vec{c}|^2}$

(D) $|\vec{a}|\vec{a} - |\vec{b}|\vec{b} + |\vec{c}|\vec{c}$

Q.3

In the figure, AB, DE and GF are parallel to each other and AD, BG and EF are parallel to each other. If $CD : CE = CG : CB = 2 : 1$, then the value of area $(\triangle AEG)$: area (ΔABD) is equal to



(A) $7/2$

(B) 3

(C) 4

(D) $9/2$

Q.4

If $\vec{p} = \frac{\vec{b} \times \vec{c}}{[\vec{a}\vec{b}\vec{c}]}, \vec{q} = \frac{\vec{c} \times \vec{a}}{[\vec{a}\vec{b}\vec{c}]}$ and $\vec{r} = \frac{\vec{a} \times \vec{b}}{[\vec{a}\vec{b}\vec{c}]}$, where $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-coplanar vectors, then the value of the expression $(\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{p} + \vec{q} + \vec{r})$ is

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(A) 3

(B) 2

(C) 1

(D) 0

Q.5

A particle moves along a curve so that its coordinates at time t are $x = t, y = \frac{1}{2}t^2, z = \frac{1}{3}t^3$. The acceleration at $t = 1$ is

(A) $\mathrm{j} + 2\mathrm{k}$

(B) $\mathrm{j} + \mathrm{k}$

(C) $2\mathrm{j} + \mathrm{k}$

(D) none of these

Q.6

If a and b are non-collinear vectors, then the value of λ for which $\mathrm{u} = (\lambda + 2)\mathrm{a} + \mathrm{b}$ and $\mathrm{v} = (1 + 4\lambda)\mathrm{a} - 2\mathrm{b}$ are collinear is

(A) $\frac{1}{2}$

(B) $-\frac{5}{6}$

(C) $\frac{3}{4}$

(D) $\frac{1}{3}$

Q.7

If $\mathrm{u}, \mathrm{v}, \mathrm{w}$ are non-coplanar vectors and p, q are real numbers, then the equality $[3\mathrm{u}p\mathrm{v}p\mathrm{w}] - [p\mathrm{v}\mathrm{w}q\mathrm{u}] - [2\mathrm{w}q\mathrm{v}q\mathrm{u}] = 0$ holds for

(A) more than two but not all values of (p, q)

(B) all values of (p, q)

(C) exactly one values of (p, q)

(D) exactly two values of (p, q)

Q.8

Two adjacent sides of a parallelogram $ABCD$ are given by $\overrightarrow{AB} = 2\hat{i} + 10\hat{j} + 11\hat{k}$ and $\overrightarrow{AD} = -\hat{i} + 2\hat{j} + 2\hat{k}$. The side AD is rotated by an acute angle α in the plane of the parallelogram so that AD becomes AD' If AD' makes a right angle with the side AB, then the cosine of the angle α is given by

(A) $\frac{8}{9}$

(B) $\frac{\sqrt{17}}{9}$

(C) $\frac{1}{9}$

(D) $\frac{4\sqrt{5}}{9}$

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Q.9

Two vectors in space are equal only if they have equal component in

(A) a given direction

(B) two given directions

(C) three given directions

(D) in any arbitrary direction

Q.10

The vectors a and b are not perpendicular and c and d are the vectors satisfying $\mathrm{b} \times \mathrm{c} = \mathrm{b} \times \mathrm{d}$ and $\mathrm{a} \cdot \mathrm{d} = 0$. Then the vector d is equal to :

(A) $\mathrm{c} - \left(\frac{\mathrm{a} \cdot \mathrm{c}}{\mathrm{a} \cdot \mathrm{b}}\right) \mathrm{b}$

(B) $\mathrm{b} - \left(\frac{\mathrm{b} \cdot \mathrm{c}}{\mathrm{a} \cdot \mathrm{b}}\right) \mathrm{c}$

(C) $\mathrm{c} + \left(\frac{\mathrm{a} \cdot \mathrm{c}}{\mathrm{a} \cdot \mathrm{b}}\right) \mathrm{b}$

(D) $\mathrm{b} + \left(\frac{\mathrm{b} \cdot \mathrm{c}}{\mathrm{a} \cdot \mathrm{b}}\right) \mathrm{c}$

Q.11

If C is the mid point of AB and P is any point outside AB, then

(A) $\mathrm{PA} + \mathrm{PB} + 2\mathrm{PC} = 0$

(B) $\mathrm{PA} + \mathrm{PB} + \mathrm{PC} = 0$

(C) $\mathrm{PA} + \mathrm{PB} = 2\mathrm{PC}$

(D) $\mathrm{PA} + \mathrm{PB} = \mathrm{PC}$

Q.12

If A, B, C, D are four points in space satisfying $\mathrm{AB} \cdot \mathrm{CD} = K\left[|\mathrm{AD}|^2 + |\mathrm{BC}|^2 - |\mathrm{AC}|^2 - |\mathrm{BD}|^2\right]$ then the value of K is

(A) 2

(B) $1/3$

(C) $1/2$

(D) 1

Q.13

For unit vectors b and c and any non zero vector a, the value of $\{\{(\mathrm{a}+\mathrm{b}) \times (\mathrm{a}+\mathrm{c})\} \times (\mathrm{b} \times \mathrm{c})\} \cdot (\mathrm{b}+\mathrm{c})$ is

(A) $|\mathrm{a}|^2$

(B) $2|\mathrm{a}|^2$

(C) $3|\mathrm{a}|^2$

(D) none of these

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Q.14

Let A B C D be a parallelogram such that $AB = \mathrm{q}$, $AD = \mathrm{p}$ and B A D be an acute angle. If r is the vector that coincides with the altitude directed from the vertex B to the side A D, then r is given by:

(A) $\mathrm{r} = -\mathrm{q} + \frac{(\mathrm{p}\cdot\mathrm{q})}{\mathrm{p}\cdot\mathrm{p}}\mathrm{p}$

(B) $\mathrm{r} = \mathrm{q} - \frac{(\mathrm{p}\cdot\mathrm{q})}{\mathrm{p}\cdot\mathrm{p}}\mathrm{p}$

(C) $\mathrm{r} = -3\mathrm{q} + \frac{3(\mathrm{p}\cdot\mathrm{q})}{\mathrm{p}\cdot\mathrm{p}}\mathrm{p}$

(D) $\mathrm{r} = 3\mathrm{q} - \frac{3(\mathrm{p}\cdot\mathrm{q})}{\mathrm{p}\cdot\mathrm{p}}\mathrm{p}$

Q.15

The edges of a parallelopiped are of unit length and parallel to non-coplanar unit length and are parallel to non-coplanar unit vectors $\mathrm{a}, \mathrm{b}, \mathrm{c}$ such that $\mathrm{a}\cdot\mathrm{b} = \mathrm{b}\cdot\mathrm{c} = \mathrm{c}\cdot\mathrm{a} = 1/2$. Then the volume of the parallelopiped is

(A) $1/\sqrt{2}$

(B) $1/2\sqrt{2}$

(C) $\sqrt{3}/2$

(D) $1/\sqrt{3}$

Q.16

Let a, b and c be distinct non-negative numbers. If the vectors $a\mathrm{i} + a\mathrm{j} + c\mathrm{k}, \mathrm{i} + \mathrm{k}$ and $c\mathrm{i} + c\mathrm{j} + b\mathrm{k}$ lie in a plane, then c is

(A) equal to zero

(B) the harmonic mean of a and b

(C) the geometric mean of a and b

(D) the arithmetic mean of a and b.

Q.17

Let A B C be a triangle whose circumcentre is at P. If the positive vectors of $\mathrm{A}, \mathrm{B}, \mathrm{C}$ and P are a, b, c and $\frac{1}{4}(\mathrm{a}+\mathrm{b}+\mathrm{c})$ respectively, then the position vector of the orthocentre of this triangle, is:

(A) $-\frac{1}{2}(\mathrm{a}+\mathrm{b}+\mathrm{c})$

(B) $\mathrm{a}+\mathrm{b}+\mathrm{c}$

(C) $\frac{1}{2}(\mathrm{a}+\mathrm{b}+\mathrm{c})$

(D) 0

Q.18

In a parallelogram $ABCD$, $|\mathrm{AB}| = a$, $|\mathrm{AD}| = b$ and $|\mathrm{AC}| = c$, then $\mathrm{DB}.\,\mathrm{AB}$ has the value: (choices modified from original)

(A) $\frac{1}{2}\left(3a^2 + b^2 - c^2\right)$

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(B) $\frac{1}{4}\left(a^2+b^2-c^2\right)$

(C) $\frac{1}{3}\left(b^2+c^2-a^2\right)$

(D) $\frac{1}{2}\left(a^2+b^2+c^2\right)$

Q.19

Let $\mathrm{p}, \mathrm{q}, \mathrm{r}$ be three mutually perpendicular vectors of the same magnitude. If a vector x satisfies the equation $\mathrm{p} \times((\mathrm{x}-\mathrm{q}) \times \mathrm{p})+\mathrm{q} \times((\mathrm{x}-\mathrm{r}) \times \mathrm{q})+\mathrm{r} \times((\mathrm{x}-\mathrm{p}) \times \mathrm{r})=0$ then x is given by

(A) $(1/2)(\mathrm{p}+\mathrm{q}-2\mathrm{r})$

(B) $(1/2)(\mathrm{p}+\mathrm{q}+\mathrm{r})$

(C) $(1/3)(\mathrm{p}+\mathrm{q}+\mathrm{r})$

(D) $(1/3)(2\mathrm{p}+\mathrm{q}-\mathrm{r})$

Q.20

Let $\mathrm{a}=\mathrm{i}+2\mathrm{j}+\mathrm{k}, \mathrm{b}=\mathrm{i}-\mathrm{j}+\mathrm{k}$ and $\mathrm{c}=\mathrm{i}+\mathrm{j}-\mathrm{k}$. A vector in the plane of a and b whose projection on c is $1/\sqrt{3}$ is

(A) $4\mathrm{i}-\mathrm{j}+4\mathrm{k}$

(B) $3\mathrm{i}+\mathrm{j}+3\mathrm{k}$

(C) $2\mathrm{i}+\mathrm{j}+2\mathrm{k}$

(D) $-4\mathrm{i}+\mathrm{j}-4\mathrm{k}$

Q.21

If $\vec{a}$ and $\vec{b}$ are any two unit vectors, then find the greatest positive integer in the range of $\frac{3|\vec{a}+\vec{b}|}{2}+2|\vec{a}-\vec{b}|$

Q.22

Let $\overrightarrow{OA}=\vec{a}, \overrightarrow{OB}=10\vec{a}+2\vec{b}$ and $\overrightarrow{OC}=\vec{b}$, where O, A and C are non-collinear points. Let p denote the area of quadrilateral $OACB$, and let q denote the area of parallelogram with OA and OC as adjacent sides. If $p=kq$, then find k.

Q.23

Let $\vec{a}=\alpha\hat{i}+2\hat{j}-3\hat{k}, \vec{b}=\hat{i}+2\alpha\hat{j}-2\hat{k}$ and $\vec{c}=2\hat{i}-\alpha\hat{j}+\hat{k}$. Find the value of 6α, such that $\{(\vec{a}\times\vec{b})\times(\vec{b}\times\vec{c})\}\times(\vec{c}\times\vec{a})=0$

Q.24

Let $\vec{u}$ and $\vec{v}$ be unit vectors such that $\vec{u}\times\vec{v}+\vec{u}=\vec{w}$ and $\vec{w}\times\vec{u}=\vec{v}$. Find the value of $[\vec{u}\vec{v}\vec{w}]$.

Q.25

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Find the value of λ if the volume of a tetrahedron whose vertices are with position vectors $\hat{i}-6\hat{j}+10\hat{k}, -\hat{i}-3\hat{j}+7\hat{k}, 5\hat{i}-\hat{j}+\lambda\hat{k}$ and $7\hat{i}-4\hat{j}+7\hat{k}$ is 11 cubic unit.

Q.26

If $\mathrm{a}, \mathrm{b}, \mathrm{c}$ are unit vectors, then the maximum value of $|\mathrm{a}+2\mathrm{b}|^2+|\mathrm{b}+3\mathrm{c}|^2+|\mathrm{c}+4\mathrm{a}|^2$ is

Q.27

If $\mathrm{a}, \mathrm{b}, \mathrm{c}$, are non-coplanar vectors such that $(2h+k)\mathrm{a}+(3-4h+l)\mathrm{b}+(1+h+k)\mathrm{c}=h\mathrm{a}+k\mathrm{b}+l\mathrm{c}$ then the value of h+k+l is

Q.28

Let a, b and c be three non-zero vectors, no two of which are collinear. If the vectors $\mathrm{a}+2\mathrm{b}$ is collinear with c, and $\mathrm{b}+3\mathrm{c}$ is collinear with a, then $\mathrm{a}+2\mathrm{b}+6\mathrm{c}$ is equal to, (λ being some non-zero scalar)

Q.29

If a, b and c are unit coplanar vectors, then the scalar triple product $[2\mathrm{a}-\mathrm{b}\ 2\mathrm{b}-\mathrm{c}\ 2\mathrm{c}-\mathrm{a}]=$

Q.30

Let $\vec{u}$ be a vector on rectangular coordinate system with sloping angle 60°. Suppose that $|\vec{u}-\hat{i}|$ is geometric mean of $|\vec{u}|$ and $|\vec{u}-2\hat{i}|$, where $\hat{i}$ is the unit vector along the x-axis. Then find the value of $(\sqrt{2}+1)|\vec{u}|$.

Answers & Solutions

Q.1 Answer:

$3\pi/4$

Solution:

We have $|\mathrm{a}.\mathrm{b}|=|\mathrm{a}\times\mathrm{b}|$
$\Rightarrow |\mathrm{a}||\mathrm{b}||\cos\theta|=|\mathrm{a}||\mathrm{b}||\sin\theta|$
(where θ is the angle between a and b) $\Rightarrow |\cos\theta|=|\sin\theta|\Rightarrow\theta=\frac{\pi}{4}$ **or** $\frac{3\pi}{4}$
But a.b $<0, \therefore\theta=\frac{3\pi}{4}$

Q.2 Answer:

$\frac{\vec{a}}{|\vec{a}|}+\frac{\vec{b}}{|\vec{b}|}+\frac{\vec{c}}{|\vec{c}|}$

Solution:

$\bar{\alpha}=\frac{\bar{a}}{|a|}+\frac{\bar{b}}{|b|}+\frac{\bar{c}}{|c|}$
$\alpha,\bar{a}\to\theta \quad \alpha,\bar{b}=\phi \quad \alpha,\bar{c}=\psi$
$\bar{\alpha}\cdot\bar{a}=\frac{\bar{a}\cdot\bar{a}}{|a|}\Rightarrow|\bar{\alpha}||a|\cos\theta=|a|$
$\cos\theta=\frac{1}{|\alpha|}$
Similarly, $\cos\phi=\frac{1}{|\alpha|}$
$\therefore\theta=\phi=\psi$

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$\cos\psi = \frac{1}{|\alpha|}$
$\alpha = \frac{\bar{a}}{|a|} + \frac{\bar{b}}{|b|} + \frac{\bar{c}}{|c|}$

Q.3 Answer:

$7/2$

Solution:

Let A **be the origin.** $\overrightarrow{\mathrm{AB}} = \overrightarrow{\mathrm{a}}, \overrightarrow{\mathrm{AD}} = \overrightarrow{\mathrm{b}}$
so, $\overrightarrow{\mathrm{AE}} = \overrightarrow{\mathrm{b}} + \frac{3}{2}\overrightarrow{\mathrm{a}}, \overrightarrow{\mathrm{AG}} = \overrightarrow{\mathrm{a}} + 3\overrightarrow{\mathrm{b}}$
So the required ratio $= \frac{\frac{1}{2}\left|(\mathrm{a}+3\vec{b})\times\left(\vec{b}+\frac{3}{2}\vec{a}\right)\right|}{\frac{1}{2}|\vec{a}\times\mathrm{b}|}$
$= \frac{7}{2}$

Q.4 Answer:

3

Solution:

Since, $\mathbf{a}\cdot\mathbf{p} = \mathbf{a}\cdot\frac{(\mathbf{b}\times\mathbf{c})}{[a\mathbf{bc}]} = \frac{[a\mathbf{b}c]}{[abc]} = 1$
Also, $\mathbf{a}\cdot\mathbf{a} = \mathbf{a}\cdot\frac{c\times a}{[abc]} = \frac{[aba]}{[abc]} = 0$
Similarly, $\mathbf{a}\cdot\mathbf{r} = 0, \quad \mathbf{b}\cdot\mathbf{p} = 0, \quad \mathbf{b}\cdot\mathbf{q} = 1, \mathbf{b}\cdot\mathbf{r} = 0$
$\mathbf{c}\cdot\mathbf{p} = 0, \quad \mathbf{c}\cdot\mathbf{q} = 0$ **and** $\mathbf{c}\cdot\mathbf{r} = 1$
$\therefore (\mathbf{a}+\mathbf{b}+\mathbf{c})\cdot(\mathbf{p}+\mathbf{q}+\mathbf{r}) = (\mathbf{a}\cdot\mathbf{p}+\mathbf{a}\cdot\mathbf{q}+\mathbf{a}\cdot\mathbf{r}) + (\mathbf{b}\cdot\mathbf{p}+\mathbf{b}\cdot\mathbf{q}+\mathbf{b}\cdot\mathbf{r}) + (\mathbf{c}\cdot\mathbf{p}+\mathbf{c}\cdot\mathbf{q}+\mathbf{c}\cdot\mathbf{r})$
$= 1+1+1 = 3$

Q.5 Answer:

$\mathrm{j}+2\mathrm{k}$

Solution:

Let $\mathrm{r} = t\mathrm{i} + \frac{1}{2}t^2\mathrm{j} + \frac{1}{3}t^3\mathrm{k}$ **then the velocity** $\mathrm{v} = \frac{\mathrm{dr}}{\mathrm{d}t} = \mathrm{i} + t\mathrm{j} + t^2\mathrm{k}$
and the acceleration $\mathrm{a} = \frac{\mathrm{dv}}{\mathrm{d}t} = \mathrm{j} + 2t\mathrm{k}.$
Hence a $\mathrm{I}_{(t=1)} = \mathrm{j}+2\mathrm{k}$

Q.6 Answer:

$-\frac{5}{6}$

Solution:

$\Rightarrow \vec{u} = \mu\vec{v} \quad \Rightarrow (\lambda+2)\vec{a} + \vec{b} = \mu(1+4\lambda)\vec{a} - 2\mu\vec{b}$
$\therefore 1 = -2\mu \& (\lambda+2) = \mu(1+4\lambda)$
$\Rightarrow \mu = -1/2 \quad \Rightarrow \lambda+2 = -1/2(1+4\lambda)$
$\Rightarrow 2\lambda+4 = -1-4\lambda$
$\Rightarrow 6\lambda = -5 \Rightarrow \lambda = \frac{-5}{6}$

Q.7 Answer:

exactly one values of (p, q)

Solution:

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$0 = [3\mathrm{u}pv\mathrm{v}p\mathrm{w}] - [p\mathrm{v}\mathrm{w}q\mathrm{u}] - [2\mathrm{w}q\mathrm{v}q\mathrm{u}]$

$= \left(3p^2 - pq + 2q^2\right)[\mathrm{uvw}]$

As u, v, w are non - coplanar so $[\mathrm{uvw}] \neq 0$**, so** $3p^2 - pq + 2q^2 = 0$

$\Rightarrow 2\left(q - \frac{1}{4}p\right)^2 + \frac{23}{8}p^2 = 0$

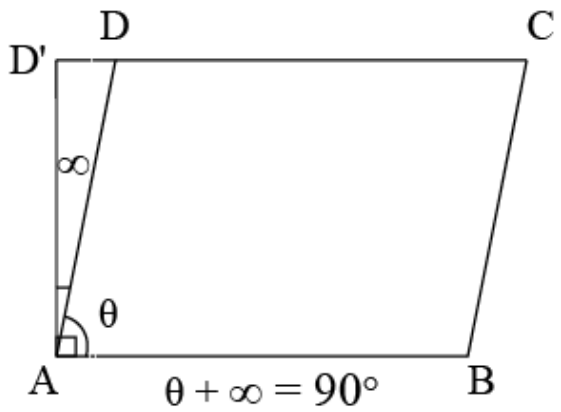
$\Rightarrow p = 0, q = \frac{1}{4}p = 0$

Thus there is exactly one value of (p, q)**.**

Q.8 Answer:

$\frac{\sqrt{17}}{9}$

Solution:



$\overline{AB} = 2\hat{i} + 10\hat{j} + 11\hat{k}$

$\overline{AD} = -\hat{i} + 2\hat{j} + 2\hat{k}$

Angle ' θ **' between** $\overrightarrow{AB}$ **and** $\overline{AD}$ **is**

$\cos(\theta) = \left|\frac{\overline{\mathrm{AB}} \cdot \overline{\mathrm{AD}}}{|\overline{\mathrm{AB}}||\overline{\mathrm{AD}}|}\right| = \left|\frac{-2+20+22}{(15)(3)}\right| = \frac{8}{9}$

$\sin\theta = \frac{\sqrt{81-64}}{9} = \frac{\sqrt{17}}{9}$

$\theta + \alpha = 90^\circ$

$\alpha = 90^\circ - \theta$

$\cos\alpha = \cos(9$

$\cos\alpha = \cos(90^\circ - \theta)$

$= \sin\theta + \cos(90^\circ - \theta)$

$= \sin\theta$

$= \frac{\sqrt{17}}{9}$

Q.9 Answer:

three given directions

Solution:

Let $\bar{a}$ **and** $\bar{b}$ **be two vectors.**

Hence

$\overline{\mathrm{a}} = \mathrm{a}_x\mathrm{i} + \mathrm{a}_y\mathrm{j} + \mathrm{a}_2\mathrm{k}$

Similarly

$\bar{b} = \mathrm{b}_x\mathrm{i} + \mathrm{b}_y\mathrm{j} + \mathrm{b}_z\mathrm{k}$

Where i, j, k **are unit vectors along** x, y **and** z **axes.**

For the vectors $\bar{a}$ **and** $\overline{\mathrm{b}}$ **to be equal, ie** $\overline{\mathrm{a}} = \overline{\mathrm{b}}$

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$a_x = b_x$

$a_y = b_y$

$a_x = b_z$

Hence answer is option C.

Q.10 Answer:

$c - \left(\frac{a \cdot c}{a \cdot b}\right) b$

Solution:

$b \times c = b \times d \Rightarrow b \times (c - d) = 0$

$\Rightarrow \quad c - d || b$

$\Rightarrow \quad c - d = \alpha b$ **for some** $\alpha \in R$

$\Rightarrow \quad d = c - \alpha b$

$\text{Also}, a \cdot d = a \cdot c - \alpha a \cdot b$

$\Rightarrow \quad 0 = a \cdot c - \alpha a \cdot b$

$\Rightarrow \quad \alpha = \frac{a \cdot c}{a \cdot b}$

Thus, $\quad d = c - \frac{a \cdot c}{a \cdot b} b.$

Q.11 Answer:

$PA + PB = 2PC$

Solution:

$\overrightarrow{PA} = \overrightarrow{PC} + \overrightarrow{CA}$

$\overrightarrow{PB} = \overrightarrow{PC} + \overrightarrow{CB}$

$\overrightarrow{PA} + \overrightarrow{PB} = 2\overrightarrow{PC} + (\overrightarrow{CA} + \overrightarrow{CB})$

$= 2\overrightarrow{PC} \quad (\because \overrightarrow{CA} = -\overrightarrow{CB})$

Hence $\overrightarrow{CA} + \overrightarrow{CB} = 0$

and $\overrightarrow{PA} + \overrightarrow{PB} = 2\overrightarrow{PC}$

Q.12 Answer:

$1/2$

Solution:

Let A **be the origin of reference and the position vector of B, C, D be** b, c, d, **w.r.t. A .**

So $AB = b, CD = d - c, AD = d, BC = c - b, AC = c$ **and** $BD = d - b.$

The L.H.S. is equal to $b \cdot (d - c).$

The R.H.S. is

\begin{array} K\left[|\mathrm{d}|^{2}+|\mathrm{c}-\mathrm{b}|^{2}-|\mathrm{c}|^{2}-|\mathrm{d}-\mathrm{b}|^{2}\right] \\= K[\mathrm{d} \cdot \mathrm{d}+\mathrm{c} \cdot \mathrm{c}+\mathrm{b} \cdot \mathrm{b}-2 \mathrm{c} \cdot \mathrm{b}-\mathrm{c} \cdot \mathrm{c}-\mathrm{d} \cdot \mathrm{d}-\mathrm{b} \cdot \mathrm{b}+2 \mathrm{d} \cdot \mathrm{b}] \\= 2 K[\mathrm{b} \cdot(\mathrm{d}-\mathrm{c}]] . \text { Hence } K=1 / 2 . \end{array}

Q.13 Answer:

none of these

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Solution:

The given expression

$$= \{\{a \times c + b \times a + b \times c\} \times (b \times c)\} \cdot (b + c)$$
$$= \{(a \times c) \times (b \times c) + (b \times a) \times (b \times c)\} \cdot (b + c)$$
$$= [(a \cdot (b \times c))c - (c \cdot (b \times c))a + (b \cdot (b \times c))a - (a \cdot (b \times c)b)] \cdot (b + c)$$
$$= [(a \cdot (b \times c))(c - b) \cdot (b + c)]$$
$$= (a \cdot (b \times c)) \left[|c|^2 - |b|^2\right] = 0$$
$$[\because |b| = |c| = 1]$$

Q.14 Answer:

$r = -q + \frac{(p \cdot q)}{p \cdot p} p$

Solution:

Let M be the foot of perpendicular from B to AD.

Suppose $AM = \alpha p$

Also, $AM = AB + BM = q + r$

As $q + r = \alpha p$, we get

$$(q + r) \cdot p = \alpha p \cdot p$$
$$\Rightarrow \quad \textbf{q.p} + r \cdot p = \alpha \textbf{ p.p}$$
$$\Rightarrow \alpha = \frac{p \cdot q}{p \cdot p} [\because r \cdot p = 0]$$

Thus, $r = -q + \frac{q \cdot p}{p \cdot p} p.$

Q.15 Answer:

$1/\sqrt{2}$

Solution:

If V is volume of the parallelopiped, then

$$V^2 = [abc]^2 = \begin{vmatrix} a \cdot a & a \cdot b & a \cdot c \\ b \cdot a & b \cdot b & b \cdot c \\ c \cdot a & c \cdot b & c \cdot c \end{vmatrix} = \begin{vmatrix} 1 & 1/2 & 1/2 \\ 1/2 & 1 & 1/2 \\ 1/2 & 1/2 & 1 \end{vmatrix} = 2 \begin{vmatrix} 1 & 1/2 & 1/2 \\ 1 & 1 & 1/2 \\ 1 & 1/2 & 1 \end{vmatrix}$$

$$[C_1 \to C_1 + C_2 + C_3]$$
$$= 2 \begin{vmatrix} 1 & 1/2 & 1/2 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/2 \end{vmatrix} = \frac{1}{2}$$
$$\Rightarrow V = 1/\sqrt{2}$$

Q.16 Answer:

the geometric mean of a and b

Solution:

$$\begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0$$
$$\Rightarrow \begin{vmatrix} 0 & a & c \\ 1 & 0 & 1 \\ 0 & c & b \end{vmatrix} = 0 \Rightarrow -ab + c^2 = 0$$
$$\Rightarrow a, b, c \textbf{ are in G.P.}$$

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Q.17 Answer:

$\frac{1}{2}(\mathrm{a}+\mathrm{b}+\mathrm{c})$

Solution:

Let p be the position vectors of the circumcentre and orthocentre be p and h.

As G divides HP in the ratio 2: 1,

$\frac{1}{3}(\mathrm{a}+\mathrm{b}+\mathrm{c})=\frac{1}{3}(2\mathrm{p}+\mathrm{h})$

$\Rightarrow \mathrm{h}=\frac{1}{2}(\mathrm{a}+\mathrm{b}+\mathrm{c})$

Q.18 Answer:

$\frac{1}{2}\left(3a^2+b^2-c^2\right)$

Solution:

$\mathrm{AC}=\mathrm{AB}+\mathrm{BC}$

$\Rightarrow \mathrm{c}=\mathrm{a}+\mathrm{b}$

and $\mathrm{DB}=\mathrm{AB}-\mathrm{AD}=\mathrm{a}-\mathrm{b}.$

Now, $\mathrm{DB}\cdot\mathrm{AB}=(\mathrm{a}-\mathrm{b})\cdot\mathrm{a}=\mathrm{a}^2-\mathrm{b}.\mathrm{a}$

Also $\mathrm{c}^2=|\mathrm{c}|^2=|\mathrm{a}+\mathrm{b}|^2=|\mathrm{a}|^2+|\mathrm{b}|^2+2\mathrm{a}\cdot\mathrm{b}$

$\Rightarrow \mathrm{a}\cdot\mathrm{b}=\frac{1}{2}\left(c^2-a^2-b^2\right)$

Thus, $\mathrm{DB}\cdot\mathrm{AB}=a^2-\frac{1}{2}\left(c^2-a^2-b^2\right)$

$=\frac{1}{2}\left(3a^2-b^2-c^2\right)$

Q.19 Answer:

$(1/2)(\mathrm{p}+\mathrm{q}+\mathrm{r})$

Solution:

Let $|\mathrm{p}|=|\mathrm{q}|=|\mathrm{r}|=K.$

Let $\mathrm{a},\mathrm{b},\mathrm{c}$ be unit vectors along $\mathrm{p},\mathrm{q},\mathrm{r}$ respectively.

Clearly $\mathrm{p},\mathrm{q},\mathrm{r}$ are mutually perpendicular vectors, so any vector x can be written as $a_1\mathrm{a}+a_2\mathrm{b}+a_3\mathrm{c}$

$\mathrm{p}\times((\mathrm{x}-\mathrm{q})\times\mathrm{p})=(\mathrm{p}\cdot\mathrm{p})(\mathrm{x}-\mathrm{q})-(\mathrm{p}\cdot(\mathrm{x}-\mathrm{q}))\mathrm{p}$

$=K^2(\mathrm{x}-\mathrm{q})-(\mathrm{p}\cdot\mathrm{x})\mathrm{p}(\mathrm{p}\cdot\mathrm{q}=0)$

$=K^2(\mathrm{x}-\mathrm{q})-K\mathrm{a}\cdot(a_1\mathrm{a}+a_2\mathrm{b}+a_3\mathrm{c})\,K\mathrm{a}$

$=K^2\left(\mathrm{x}-\mathrm{q}-a_1\mathrm{a}\right)$

Similarly $\mathrm{q}\times((\mathrm{x}-\mathrm{r})\times\mathrm{q})=K^2\left(\mathrm{x}-\mathrm{r}-a_2\mathrm{b}\right)$

and $\mathrm{r}\times((\mathrm{x}-\mathrm{p})\times\mathrm{r})=K^2\left(\mathrm{x}-\mathrm{p}-a_3\mathrm{c}\right)$

According to the given condition $K^2\left(\mathrm{x}-\mathrm{q}-a_1\mathrm{a}+\mathrm{x}-\mathrm{r}-a_2\mathrm{b}+\mathrm{x}-\mathrm{p}-a_3\mathrm{c}\right)=0$

$\Rightarrow \quad\{3\mathrm{x}-(\mathrm{p}+\mathrm{q}+\mathrm{r})-(a_1\mathrm{a}+a_2\mathrm{b}+a_3\mathrm{c})\}=0$

$\Rightarrow \quad[2\mathrm{x}-(\mathrm{p}+\mathrm{q}+\mathrm{r})]=0$

$\Rightarrow x=(1/2)(\mathrm{p}+\mathrm{q}+\mathrm{r}).$

Q.20 Answer:

$-4\mathrm{i}+\mathrm{j}-4\mathrm{k}$

Solution:

A vector d in the plane of a and b is given by, $d=\lambda a+\mu b=(\lambda+\mu)i+(2\lambda-\mu)j+(\lambda+\mu)\mathrm{k}$

Now the projection of don c is $\frac{\mathrm{c}\cdot\mathrm{d}}{|\mathrm{c}|}=\frac{\lambda+\mu+(2\lambda-\mu)-(\lambda+\mu)}{\sqrt{3}}=\frac{1}{\sqrt{3}}$ (given) $\Rightarrow 2\lambda-\mu=1$. Thus $\mathrm{d}=(3\lambda-$

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$1)\mathrm{i}+\mathrm{j}+(3\lambda-1)\mathrm{k}$,
So for $\lambda=-1, d=-4i+j-4k$

Q.21 Answer:

—

Solution:

Let angle between $\bar{a}$ **and** $\vec{b}$ **be** θ
We have $|\overrightarrow{\mathrm{a}}|=|\overrightarrow{\mathrm{b}}|=1$
Now $|\overrightarrow{\mathrm{a}}+\overrightarrow{\mathrm{b}}|=2\cos\frac{\theta}{2}$ **and** $|\overrightarrow{\mathrm{a}}-\overrightarrow{\mathrm{b}}|=2\sin\frac{\theta}{2}$
Consider $\mathrm{F}(\theta)=\frac{3}{2}\left(2\cos\frac{\theta}{2}\right)+2\left(2\sin\frac{\theta}{2}\right)$
$\therefore \mathrm{F}(\theta)=3\cos\frac{\theta}{2}+4\sin\frac{\theta}{2}, \theta\in[0,\pi]$

Q.22 Answer:

—

Solution:

We have,
$\mathrm{p}=$ **Area of the quadrilateral** OABC
$=\frac{1}{2}|\overrightarrow{\mathrm{OB}}\times\overrightarrow{\mathrm{AC}}|$
$=\frac{1}{2}|\overrightarrow{\mathrm{OB}}\times(\overrightarrow{\mathrm{OC}}-\overrightarrow{\mathrm{OA}})|$
$=\frac{1}{2}|(10\vec{a}+2\vec{b})\times(\vec{b}-\vec{a})|$
$=\frac{1}{2}\mid 10(\overrightarrow{\mathrm{a}}\times\overrightarrow{\mathrm{b}}-10(\overrightarrow{\mathrm{a}}\times\overrightarrow{\mathrm{a}})+2(\overrightarrow{\mathrm{b}}\times\overrightarrow{\mathrm{b}})-2(\overrightarrow{\mathrm{b}}\times\overrightarrow{\mathrm{a}})\mid$
$=\frac{1}{2}|10(\vec{a}\times\vec{b})-0+0+2(\vec{a}\times\vec{b})|$
and, q **= Area of the parallelogram with OA and OC as adjacent sides**
$=|\overrightarrow{\mathrm{OA}}\times\overrightarrow{\mathrm{OC}}|=|\overrightarrow{\mathrm{a}}\times\overrightarrow{\mathrm{b}}|\cdots\cdots\ldots(\mathrm{ii})$
From (i) and (ii), we get $p=6q$

Q.23 Answer:

—

Solution:

$\{(\overrightarrow{\mathrm{a}}\times\overrightarrow{\mathrm{b}})\times(\overrightarrow{\mathrm{b}}\times\overrightarrow{\mathrm{c}})\}\times(\overrightarrow{\mathrm{c}}\times\overrightarrow{\mathrm{a}})$
$=[\vec{a}\vec{b}\vec{c}]\vec{b}\times(\vec{c}\times\vec{a})$
$=[\vec{a}\vec{b}\vec{c}]\{(\vec{a}\cdot\vec{b})\vec{c}-(\vec{b}\cdot\vec{c})\vec{a}\}$
which vanishes if
(i) $(\vec{a}\cdot\vec{b})\vec{c}=(\vec{b}\cdot\vec{c})\vec{a}$
(ii) $[\overrightarrow{\mathrm{a}}\,\overrightarrow{\mathrm{b}}\,\overrightarrow{\mathrm{c}}]=0$
(i) $(\vec{a}\cdot\vec{b})\vec{c}=(\vec{b}\cdot\vec{c})\vec{a}$
leads to the equation $2a^3+10\alpha+12=0, a^2+6a=0$ **and** $6\alpha-6=0$**, which do not have a common solution.**
(ii) $\left[\begin{array}{lll}\overrightarrow{\mathrm{a}} & \overrightarrow{\mathrm{b}} & \overrightarrow{\mathrm{c}}\end{array}\right]=0$

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$$\Rightarrow \begin{vmatrix} \alpha & 2 & -3 \\ 1 & 2\alpha & -2 \\ 2 & -\alpha & 1 \end{vmatrix} = 0$$

$\Rightarrow 3\alpha = 2$

$\Rightarrow \alpha = \frac{2}{3}$

$\vec{a} \cdot \vec{b} = 6,\ b \cdot \vec{c} = 0$

and the vector product is $-60(2\hat{i} + \hat{k})$.

Q.24 Answer:

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Solution:

Given that, $u \times v + u = w$

Multiplying both the sides with vector u

$$\begin{aligned} (u \times v + u) \times u &= w \times u \\ (u \times v) \times u + u \times u &= v \\ (u \cdot u)v - (v \cdot u)u + u \times u &= v \\ (1)v - (v \cdot u)u + 0 &= v \\ v - (v \cdot u)u &= v \\ u \cdot v &= 0 \end{aligned}$$

Now,

$$\begin{aligned} & u \cdot (v \times w) \\ &= u \cdot (v \times (u \times v + u)) \\ &= u \cdot (v \times (u \times v) + v \times u) \\ &= u \cdot ((v \cdot v)u - (v \cdot u)v + v \times u) \\ &= u \left(|v|^2 u - 0 + v \times u \right) \\ &= |v|^2 (u \cdot u) - u \cdot (v \times u) \\ &= |v|^2 |u|^2 - 0 \\ &= 1 \end{aligned}$$

Therefore, $[uvw] = 1$

Q.25 Answer:

—

Solution:

Let the vertices be A, B, C, D **and** O **be the origin.**

$\therefore \overrightarrow{OA} = i - 6j + 10\hat{k}, OB = i - 3j + 7\hat{k}$

$\overrightarrow{OC} = -5i - j + \lambda k, OD = 7i - 4\hat{j} + 7\mathrm{k}$

$\therefore AB = OB - \overrightarrow{OA} = -2i + 3j - 3\hat{k}$

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$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = 4\hat{i} + 5] + (\lambda - 10)\mathrm{k}$

$\mathrm{AD} = \mathrm{OD} - \overrightarrow{OA} = 6\mathrm{i} + 2\mathrm{j} - 3\mathrm{k}$

Volume of tetrahedron $\mathrm{V} = \frac{1}{6}\left[\begin{array}{ccc} \mathrm{AB} & \mathrm{AC} & \mathrm{AD} \end{array}\right] = \frac{1}{6}\begin{vmatrix} -2 & 3 & -3 \\ 4 & 5 & \lambda - 10 \\ 6 & 2 & -3 \end{vmatrix}$

$\mathrm{V} = \frac{1}{6}\{-2(-15 - 2\lambda + 20) - 3(-12 - 6\lambda + 60) - 3(8 - 10)\}$

$\mathrm{V} = \frac{1}{6}\{4\lambda - 10 - 144 + 18\lambda + 66\}$

$\mathrm{V} = \frac{1}{6}(22\lambda - 88) = 11$ **(given)**

or $2\lambda - 8 = 6$

or $\lambda = 7$

Q.26 Answer:

—

Solution:

$|\mathrm{a} + 2\mathrm{b}|^2 + |\mathrm{b} + 3\mathrm{c}|^2 + |\mathrm{c} + 4\mathrm{a}|^2$

$= |\mathrm{a}|^2 + 4|\mathrm{b}|^2 + 2 \times 2\mathrm{a} \cdot \mathrm{b} + |\mathrm{b}|^2 + 9|\mathrm{c}|^2 + 6\mathrm{b} \cdot \mathrm{c}$

$+|\mathrm{c}|^2 + 16|\mathrm{a}|^2 + 8\mathrm{a} \cdot \mathrm{c}$

$\leq 1 + 4 + 4 + 1 + 9 + 6 + 1 + 16 + 8 = 50$

Q.27 Answer:

—

Solution:

$h = 2h + k, 3 - 4h + l = k, 1 + h + k = l$

$\Rightarrow k = -h$ **so** $3 - 3h + l = 0, l = 1$

$\Rightarrow h = 4/3, k = -4/3, l = 1$

Q.28 Answer:

—

Solution:

Let $\mathrm{a} + 2\mathrm{b} = x\mathrm{c}$ **and** $\mathrm{b} + 3\mathrm{c} = y\mathrm{a}$.

Then $\mathrm{a} + 2\mathrm{b} + 6\mathrm{c} = (x + 6)\mathrm{c}$ **and**

also, $\mathrm{a} + 2\mathrm{b} + 6\mathrm{c} = (1 + 2y)\mathrm{a}$.

So $(x + 6)\mathrm{c} = (1 + 2y)\mathrm{a}$.

Since a **and** c **are non-zero and non-collinear, we have** $x + 6 = 0$ **and** $1 + 2y = 0$, **i.e.,** $x = -6$ **and** $y = -1/2$.

In either case, we have $\mathrm{a} + 2\mathrm{b} + 6\mathrm{c} = 0$.

Q.29 Answer:

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Solution:

Given [a b c] $= 0$

Now, $[2a - b2b - c2c - a]$

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$= (2a - b) \cdot \{(2b - c) \times (2c - a)\}$
$= (2a - b) \cdot \{4(b \times c) - 2(b \times a) + (c \times a)\}$
$= 8(a \cdot (b \times c)) - (b \cdot (c \times a))$
$= 8[abc] - [abc]$
$= 7[abc]$
$= 0$

Q.30 Answer:

—

Solution:

since angle between $\vec{u}$ and $\hat{i}$ is 60°
$\overrightarrow{\mathrm{u}} \cdot \mathrm{i} = |\overrightarrow{\mathrm{u}}||\hat{i}| \cos 60^\circ = \frac{|\vec{u}|}{2}$
Given that $|\overrightarrow{\mathrm{u}} - \hat{i}|, |\vec{u}|, |\overrightarrow{\mathrm{u}} - 2\hat{i}|$ are in G.P., so
$|\overrightarrow{\mathrm{u}} - \hat{\mathrm{i}}|^2 = |\overrightarrow{\mathrm{u}}||\overrightarrow{\mathrm{u}} - 2\hat{\mathrm{i}}|$
Squaring both sides, $\left[|\vec{u}|^2 + |\hat{i}|^2 - 2\vec{u} \cdot \hat{i}\right]^2 = |\vec{u}|^2 \left[|\vec{u}|^2 + 4|\hat{i}|^2 - 4\vec{u} \cdot \hat{i}\right]$
$\left[|\vec{u}|^2 + 1 - \frac{2|\vec{u}|}{2}\right]^2 = |\vec{u}|^2 \left[|\vec{u}|^2 + 4 - 4\frac{|\vec{u}|}{2}\right] \Rightarrow |\vec{u}|^2 + 2|\vec{u}| - 1 = 0 \Rightarrow |\vec{u}| = -\frac{2 \pm 2\sqrt{2}}{2} \Rightarrow |\vec{u}| = \sqrt{2} - 1$

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